Jian Guan

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Research Interests

My research focuses on **stratospheric chemistry**, its **interactions with climate**, and the **detection and attribution of atmospheric change**. I combine satellite observations, chemistry-climate models, and statistical methods to study **ozone-layer change**, **stratospheric water vapor**, and **wildfire aerosols**, with the goal of quantifying how changes in **stratospheric composition** influence **long-term climate change**, and vice versa.

Education

Massachusetts Institute of Technology, Cambridge, MA Sep 2022 – May 2027 (expected)
Ph.D. candidate in Climate Science
Advisor: Susan Solomon

Renmin University of China, Beijing, China Sep 2018 – May 2022
B.S. in Environmental Engineering

Publications

Manuscripts Submitted and Under Review

11. **Jian Guan** and Susan Solomon. (2026). A 35-year Satellite-Constrained Estimate of Stratospheric Water Vapor Radiative Forcing from Methane Oxidation. *Geophysical Research Letters* (Submitted)

10. Qindan Zhu, Arlene M. Fiore, Robert Pincus, **Jian Guan**, Clare E. Singer, Brian Medeiros, and Paolo Giani. (2026). Weak OH Response to Idealized ENSO. *Geophysical Research Letters* (In Revision)

Peer-reviewed

9. **Jian Guan**, Benjamin D. Santer, Peidong Wang, Qiang Fu, Rolando R. Garcia, Yaowei Li, Kane Stone, Douglas Kinnison, Jun Zhang, Gabriel Chiodo, Jean-Francois Lamarque, and Susan Solomon. (2026). The Emergence of Human Influence on the Ozone Layer by the 1960s. *Proceedings of the National Academy of Sciences*, 123(28), e2608286123. [Link] [Nature Research Highlight] [MIT news]

8. Qindan Zhu, Nicole Neumann, Arlene M. Fiore, Robert Pincus, **Jian Guan**, George Milly, Clare Singer, Brian Medeiros, and Paolo Giani. (2026). Uncertain Natural Emissions Dampen the Increase in Tropospheric Hydroxyl Radical (OH) With Idealized Surface Warming. *Journal of Advances in Modeling Earth Systems*, 18(3), e2025MS005248. [Link] [MIT news]

7. **Jian Guan**, Susan Solomon, Daniel M. Murphy, Kane Stone, Pengfei Yu, Douglas Kinnison, Gregory P. Schill, Simone Tilmes, and Michael J. Lawler. (2025). Limited Long-Term Photolysis of Stratospheric Organic Aerosols With Implications for CESM Modeling. *Journal of Advances in Modeling Earth Systems*, 17(11), e2025MS005084. [Link]

6. Kane Stone, Susan Solomon, Pengfei Yu, Daniel M. Murphy, Douglas Kinnison, and **Jian Guan**. (2025). Two-years of stratospheric chemistry perturbations from the 2019/2020 Australian wildfire

smoke. *Atmospheric Chemistry and Physics*, 25(14), 7683–7697. [Link]

5. Jun Zhang, Peidong Wang, Douglas Kinnison, Susan Solomon, **Jian Guan**, Kane Stone, and Yunqian Zhu. (2024). Stratospheric chlorine processing after the unprecedented Hunga Tonga eruption. *Geophysical Research Letters*, 51(17), e2024GL108649. [Link]

4. **Jian Guan**, Susan Solomon, Sasha Madronich, and Douglas Kinnison. (2023). Inferring the photolysis rate of NO_2 in the stratosphere based on satellite observations. *Atmospheric Chemistry and Physics*, 23(18), 10413–10422. [Link]

3. Lei Wang, **Jian Guan** (co-first author), Hao Han, Mingyue Yao, Jian Kang, Meng Peng, Desheng Wang, Jiayu Xu, and Jiming Hao. (2022). Enhanced photocatalytic removal of ozone by a new chlorine-radical-mediated strategy. *Applied Catalysis B: Environmental*, 306, 121130. [Link]

2. **Jian Guan**, Zeqing Long, Qiangang Li, Jinchi Han, Hongbiao Du, Pengfei Wang, and Guangming Zhang. (2021). Citric acid modulated preparation of CdS photocatalyst for efficient removal of Cr(VI) and methyl orange. *Optical Materials*, 121, 111604. [Link]

1. **Jian Guan**, Bohan Jin, Yizhe Ding, Wen Wang, Guoxiang Li, and Pubu Ciren. (2021). Global surface HCHO distribution derived from satellite observations with neural networks technique. *Remote Sensing*, 13(20), 4055. [Link]

Presentations

Invited Talks

- Advancing the Understanding of Stratospheric Chemistry: Photolysis, Wildfires, and Fingerprinting. **Yuan Wang Group Meeting, Stanford University**, California, 2026.
- Limited Long-Term Photolysis of Stratospheric Organic Aerosols With Implications for CESM Modeling. **Jinan University**, Guangzhou, 2025.

Conference presentations

- Limited Long-Term Photolysis of Stratospheric Organic Aerosols With Implications for CESM Modeling. **CESM Workshop**, Colorado. 2026.
- Human Influence on the Ozone Layer Detectable by the 1960s. **Congress of China Geodesy and Geophysics**, Shanghai. 2026.
- Human Influence on the Ozone Layer Detectable by the 1960s. **American Geophysical Union Fall Meeting**, New Orleans, Louisiana. 2025.
- Human Influence on the Ozone Layer Detectable by the 1960s. **NDACC Symposium**, Virginia Beach, Virginia. 2025.
- Limited Long-Term Photolysis of Stratospheric Organic Aerosols With Implications for CESM Modeling. **Quadrennial Ozone Symposium, Boulder**, Colorado. 2024.
- The influence of stratocumulus on climate sensitivity. **Cloud Feedback Model Intercomparison Project**, Boston, Massachusetts. 2024.
- Inferring the photolysis rate of NO_2 in the stratosphere based on satellite observations. **American Geophysical Union Fall Meeting**, San Francisco, California. 2023.

Mentoring

An Phan, undergraduate researcher (MIT UROP) 2026 – Present
Project: detection and attribution of long-term polar vortex trends

Selected Honors, Awards, and Fellowships

Praecis Presidential Fellow, MIT 2022 – 2023
Jule Charney Prize, MIT 2022

Professional Service

Reviewer: Atmospheric Chemistry and Physics; Eco-Environment & Health